Chenyang Li

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Personal Information

- **Date of Birth:** October 4, 1999.
- **Nationality:** China.

Research Interests

Finite element methods, Numarical analysis, Numerical solution of PDEs.
Numerical analysis and simulation of nonlinear coupled PDEs, including Navier–Stokes, natural-convection, MHD, Stokes–Darcy, phase-field, chemotaxis–Navier–Stokes, and tumor-growth models, using backward Euler, BDF2, Crank–Nicolson, projection, Gauge–Uzawa, and SAV methods.

Technical skills

- **Programming:** FreeFem++, Fenics, TecPlot, Paraview.
- **Writing:** Research manuscripts, funding proposals.

Education

- **East China Normal University** *Sept. 2023-Present*
  Ph.D. student in Computational Mathematics. Shanghai, China
  - **Concentration:** Numerical analysis and simulation of incompressible flow coupled with multi-physics fields.
  - **Supervisor:** Haibiao Zheng, Professor, School of Mathematical Sciences and Shanghai Key Laboratory of Pure Mathematics and Mathematical Practice, East China Normal University, Shanghai, China.
- **Wenzhou University** *Sept. 2020-Jul. 2023*
  M.S. in Computational Mathematics. Wenzhou, China
  - **Concentration:** Finite element discretizations for incompressible flow with variable density.
  - **Supervisor:** Yuan Li (Associate Professor) &. Rong An (Professor). College of Mathematics and Physics, Wenzhou University, Wenzhou, China.
- **Zhejiang Ocean University** *Sept. 2016-Jul. 2020*
  B.S. in Mathematics and Applied Mathematics (Normal Major). Zhoushan, China

Experience

- **University of Dundee** *15.Sept. 2025-15.Sept. 2026*
  Associate Staff in School of Science and Engineering. Dundee, UK
  - **Concentration:** Numerical analysis and simulation of Phase-Field Models.
  - **Host:** Ping Lin, Professor, School of Science and Engineering, University of Dundee, UK.
- **Xinjiang University** *Aug. 2025*
  Academic visitor in College of Mathematics and System Sciences Urumqi China
  - **Host:** Jianping Zhao, Professor, College of Mathematics and System Sciences, Xinjiang University, Urumqi, China.

Honors and Awards

• CSC Scholarship, China Scholarship Council, China, 2025.

• Graduate Academic Scholarship, East China Normal University, Shanghai, China. 2023-2024

• Outstanding Graduates of Zhejiang Province, Wenzhou, China. 2023. June.

• Outstanding Graduates of Zhejiang Ocean University, Zhoushan, China. 2020. June.

Research Experience

[1] Algorithm study of the incompressible magnetohydrodynamic equations with variable density in 2D. Xinmiao Talents Program of Zhejiang Province, **Principal Investigator (P.I.)**, Fiscal Year 2022-2024.

[2] Convergence analysis of finite element discrete scheme for the incompressible magnetohydrodynamics system with variable density. the Master's Innovation Foundation of Wenzhou University. **Principal Investigator (P.I.)**, Fiscal Year 2022-2023.

[3] Error analysis of first-order Euler linearized finite element scheme for the 2D magnetohydrodynamics system with variable density. The Innovation Foundation of Wangxiaoan in Wenzhou University, **Principal Investigator (P.I.)**, Fiscal Year 2022-2023.

[4] Blow up and Existence of the solutions for biological chemotaxis models. The Innovation Foundation of Zhejiang Ocean University. **Principal Investigator (P.I.)**, Fiscal Year 2018-2019.

Ongoing Works

[1] **Chenyang Li**, Ping Lin, Haibiao Zheng. Unconditionally stable Gauge–Uzawa finite element schemes for the chemo-repulsion Navier-Stokes system. arXiv.2510.27026.

[2] **Chenyang Li**, Ping Lin,Hui Yu, Haibiao Zheng. A Thermodynamically Consistent Cahn−Hilliard Navier−Stokes Models For Tumor Growth. arXiv:2608.06099.

[3] **Chenyang Li**,Haibiao Zheng. A Coupled Finite-Element and Local Spherical Fokker−Planck Method for Gyrotactic Bioconvection.

[4] **Chenyang Li**, Yizhong Sun, Haibiao Zheng. Error estimate of parallel decoupled stabilized finite element algorithm for the fully mixed Stokes-Darcy Problems.

Publications

[1] **Chenyang Li**, Ping Lin, Haibiao Zheng. Fully discrete finite element approximation for the projection method to solve the Chemotaxis-Fluid System. Communications in Nonlinear Science and Numerical Simulation, 161(2026): 110099.

[2] **Chenyang Li***, Yuze Lu, Haibiao Zheng. Error Estimate of a linearized Second-order Fully Discrete Finite Element Method for the bioconvection flows with concentration dependent viscosity. Journal of computational mathematics, Accepted.

[3] **Chenyang Li**, Haibiao Zheng. Temporal error analysis of a BDF2 time-discrete scheme for the incompressible Navier-Stokes equations with variable density. Journal of Computational and Applied Mathematics 474 (2026): 0377-0427.

[4] Atrout Sabah, Md. Abdullah Al Mahbub, **Chenyang Li**, and Haibiao Zheng. Efficient and Long-Time Accurate Second-Order Decoupled Method for the Blood Solute Dynamics Model. International Journal of Numerical Analysis and Modeling. 23.1 (2026): 24-62.

[5] **Chenyang Li**, Yuan Li. Optimal L2 error analysis of first-order Euler linearized finite element scheme for the 2D magnetohydrodynamics system with variable density. Computers and Mathematics with Applications 128 (2022): 96-107.

[6] **Chenyang Li**, Jian Sun, Hailiang Zhang. Introduction of Several Biological Population Models. Hans Journal of Computational Biology. 10.12677/HJCB.2019.94010.

REFERENCES

1. **Rong An**
   Professor, College of Mathematics and Physics, Wenzhou University, Wenzhou 325035, China.
   Email: anrong@wzu.edu.cn
   Relationship: M. S. Advisor.

2. **Haibiao Zheng**
   Professor, School of Mathematical Sciences and Shanghai Key Laboratory of Pure Mathematics and Mathematical Practice, East China Normal University, Shanghai 200241,China.
   Email: hbzheng@math.ecnu.edu.cn
   Relationship: Ph.D. Advisor.

3. **Ping Lin**
   Chair (Professor) of Numerical Analysis/Computational Math (2007 –now), School of Science and Engineering, University of Dundee, UK.
   Email: p.lin@dundee.ac.uk
   Relationship: Host in School of Science & Engineering, University of Dundee.